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1. Introduction

Convolution is a core concept in signal processing. It is widely used to determine the output of a linear time-invariant (LTI) system given an input signal and its impulse response. This project focuses on developing a MATLAB App Designer-based GUI to help users visualise and understand the convolution process for both discrete-time (DT) and continuous-time (CT) signals interactively. The tool also supports cross-correlation for DT signals and animated visualization.

2. Objectives

This GUI application allows users to:

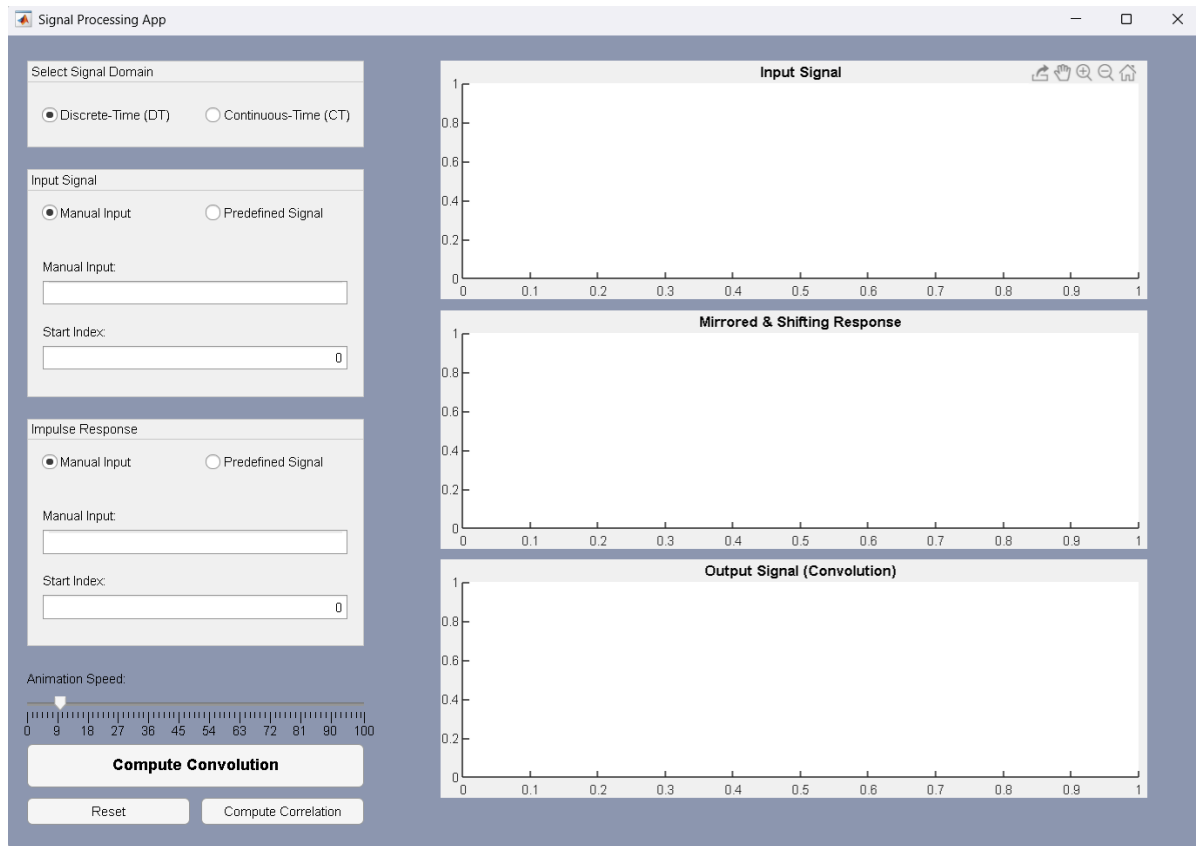
1. Select signal domain: Discrete-Time or Continuous-Time.
2. Enter or choose:
 - a. DT signals via manual vector input and starting indices.
 - b. CT signals via dropdowns for standard waveforms (impulse, step, rectangular, triangular, sawtooth) with start time, end time, and amplitude.
3. Compute and visualize the convolution process step-by-step.
4. Display three plots:
 - a. The fixed input signal
 - b. The mirrored & shifting impulse response
 - c. The real-time convolution output being built.
5. Compute correlation (for DT signals).
6. Control animation speed and reset inputs using intuitive controls.

3. Methodology

The application begins by accepting inputs from the user. In discrete-time mode, users manually enter the signal vectors for the input $x[n]$ and the impulse response $h[n]$, along with their starting indices. These vectors are processed to compute the convolution using MATLAB's built-in **conv** function. For continuous-time mode, users choose predefined waveforms and input the parameters including time range and amplitude. These waveforms are then sampled at a high frequency (typically 1000 Hz) to approximate continuous signals as dense arrays. The continuous convolution is performed using discrete convolution scaled by the time step ($\text{conv}(x,h)/fs$) to approximate the integral.

To animate the convolution process, the flipped impulse response is shifted step-by-step over the input signal. At each step, the overlapping values are multiplied and summed to generate a new output value. The GUI presents three synchronized plots: the original input signal, the mirrored and shifting impulse response. The output signal is built point by point. Users can control the animation speed. In DT mode, users also have the option to compute the cross-correlation between the two signals using MATLAB's **xcorr** function. All plots are automatically updated to reflect each step in real time, providing a clear and educational animation of the convolution sum.

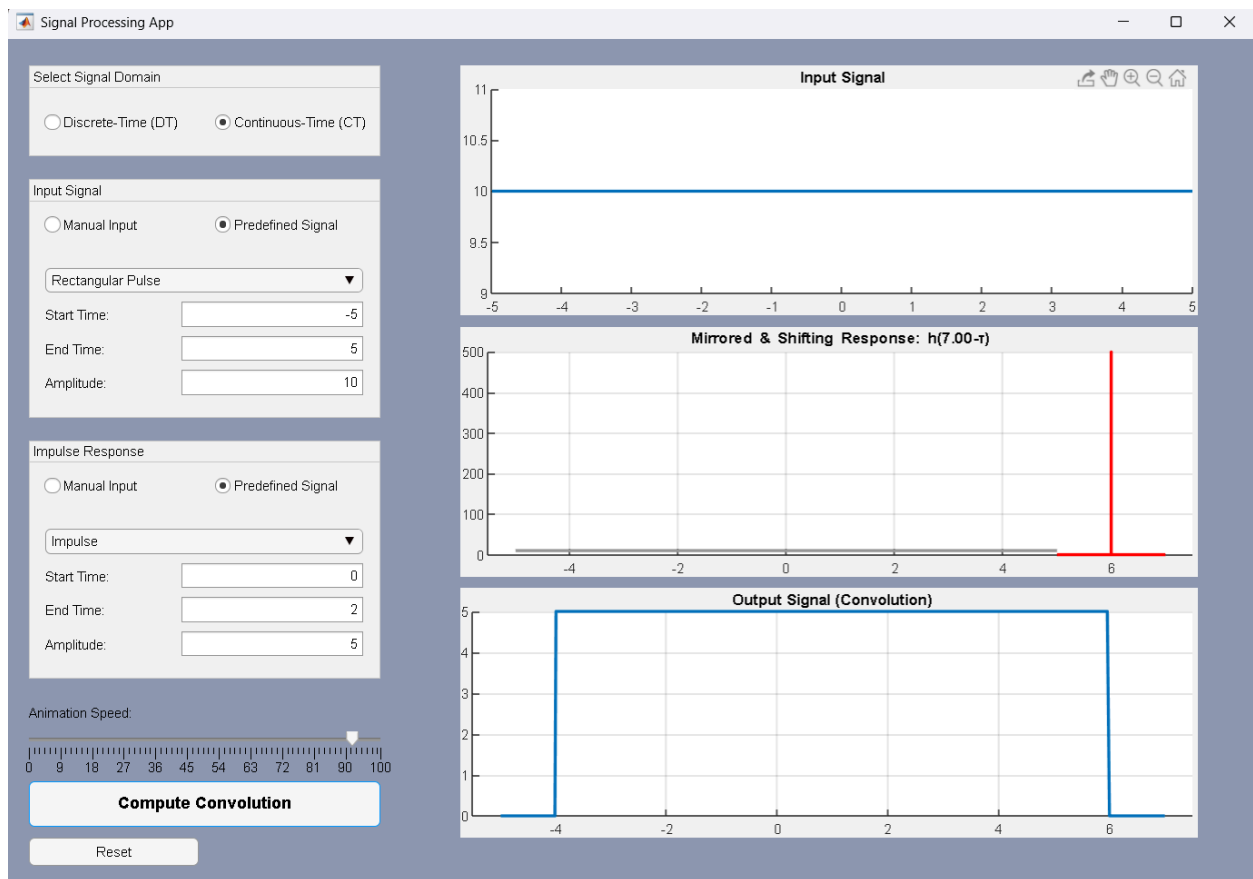
4. Design



The entire GUI was developed using MATLAB App Designer, which offers a rich set of tools for interface creation and user interaction. The main window is divided into two parts: the left panel includes all control components such as input modes, signal type selection, vector fields, time fields, dropdown menus, and buttons. The right panel consists of three axes for plotting the input signal, the mirrored and shifting impulse response, and the output signal.

For signal entry, DT mode uses editable fields where users can enter vectors and start indices, while CT mode provides dropdowns for predefined signals and numeric fields for time and amplitude. Radio buttons allow switching between DT and CT modes, and between manual or predefined input types. Buttons such as "Compute Convolution", "Reset", and "Compute Correlation" are clearly labeled and placed for ease of use. The animation speed is controlled by a slider, which dynamically adjusts the delay between frames in the animation loop.

Visual representation of convolution sum for CT signals



5. Features:

Select Signal Domain

☒ Discrete-Time (DT) ☐ Continuous-Time (CT)

Initially select whether the signal is DT or CT.

Both the input signal and the impulse signal can be manual input or predefined, and under each selection, there are input specs for different parameters.

Finally, there is the convolution button, the animation speed can be controlled with a slider and correlation for DT.

The drive link containing the video explaining the code is as follows:

<https://drive.google.com/drive/folders/1lylNICwigoFW57X-uP25oTBM58WHT-jA?usp=sharing>

6. Problems Faced

Throughout the development of the GUI, we encountered some challenges. One major issue was correctly handling signal alignment during convolution. When using different start indices or vector lengths. Managing continuous-time signals was also tricky, as we had to carefully choose the sampling rate to balance accuracy and performance.

Animating the convolution process in real time posed difficulties in synchronization across plots, and required careful tuning to avoid lag or flickering. We also faced challenges with dynamic UI behavior—ensuring that the correct input fields appeared or disappeared based on the selected signal type and input mode.

Finally, adding the correlation feature required separate logic and careful plotting to avoid confusion with convolution. These challenges helped us strengthen both our signal processing understanding and our MATLAB GUI development skills.

7. Contributions

ID	Contribution
200021357	Functions to toggle
210021313	Convolution and Correlation Part
210021321	GUI Layout and Design
210021325	Animation of the system, reset button pushed and generating the CT signal

Everyone took part in preparing documentation and reports.

8. Timeline:

Week	Task
Week 1	Project announcement and Team formation
Week 2	Learning MATLAB functions for designing the convolution sum, determining key features and input/output specifications

Week 3,4	Developed discrete time input panels and logics. Implemented manual convolution and output plotting. Added continuous time signal generation and integrated standard waveforms(impulse, step, rectangular, triangular etc)
Week 5,6	Mid exam
Week 7	Built reset and Compute buttons and ensured smooth user experience and robustness. Integrated bonus features like correlation animation speed and tested the process.
Week 8	Presentation slide, presentation and report

9. Conclusion:

This project successfully met its objective of building an interactive MATLAB GUI to visualize the convolution and correlation of discrete-time and continuous-time signals. In the process, the team gained hands-on experience with MATLAB App Designer, signal processing logic, and GUI development. The final tool serves not only as a demonstration project but also as a practical learning aid.